

ADVANCED QUANTUM MECHANICS

Exercise 7 – presentation session – 25.11.2025



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1. Plane waves, flux, currents and cross sections

1A: Derivation of the probability current in one dimension

Consider a particle moving along the z -axis, described by the time-dependent Schrödinger equation

$$i\hbar \frac{\partial \psi(z, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(z, t)}{\partial z^2} + V(z) \psi(z, t).$$

Define the probability density

$$\rho(z, t) = |\psi(z, t)|^2 = \psi^*(z, t) \psi(z, t),$$

and recall that probability must satisfy a continuity equation,

$$\frac{\partial \rho}{\partial t} + \frac{\partial j}{\partial z} = 0.$$

- (i) Write down the complex conjugate of the Schrödinger equation and compute the time derivative of the density,

$$\frac{\partial \rho}{\partial t} = \psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t}.$$

- (ii) Insert $\partial_t \psi$ and $\partial_t \psi^*$ from the Schrödinger equation and its conjugate. Show that all terms containing the potential $V(z)$ cancel, and reduce the expression to

$$\frac{\partial \rho}{\partial t} = \frac{\hbar}{2im} \left(\psi \frac{\partial^2 \psi^*}{\partial z^2} - \psi^* \frac{\partial^2 \psi}{\partial z^2} \right).$$

- (iii) Show that the expression in parentheses can be written as a total derivative:

$$\psi \psi^{*''} - \psi^* \psi'' = -\frac{\partial}{\partial z} \left(\psi^* \psi' - \psi'^* \psi \right),$$

where primes denote derivatives with respect to z .

- (iv) Insert this result into the continuity equation and identify the probability current $j(z, t)$. Conclude that

$$j(z, t) = \frac{\hbar}{2mi} \left(\psi^* \partial_z \psi - (\partial_z \psi^*) \psi \right).$$

1B: Current, flux and cross section in three dimensions

Now we extend to 3 dimensions. Consider a free particle with

$$\psi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{L^3}} e^{i\mathbf{k} \cdot \mathbf{r}}, \quad E_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m}.$$

- (i) Compute the probability current density

$$\mathbf{j}(\mathbf{r}) = \frac{\hbar}{2mi} (\psi^* \nabla \psi - (\nabla \psi^*) \psi)$$

for this plane wave and show that its magnitude is $|\mathbf{j}| = v/L^3$, where $v = \frac{\hbar k}{m}$.

- (ii) Reproduce the argument from the lecture: consider the number of particles inside a slab of thickness $v \Delta t$ and cross-sectional area L^2 . Using a uniform density $1/L^3$, show that the incident flux

$$\Phi = \frac{1}{A} \frac{\Delta N_{\text{incoming}}}{\Delta t}$$

equals $|\mathbf{j}|$ when $A = L^2$.

- (iii) A small detector with area ΔA is placed at distance r from the scatterer and oriented at polar angles (θ, ϕ) , so that it subtends a solid angle $\Delta\Omega \simeq \Delta A/r^2$. Explain how the differential cross section

$$\frac{d\sigma}{d\Omega} = \frac{1}{\Phi} \frac{\Delta N_{\text{scatt}}}{\Delta t \Delta\Omega}$$

can be inferred from counting the number of scattered particles per unit time at the detector.

2. Lippmann–Schwinger in a finite momentum basis

Consider a set of N momentum eigenstates $\{|\mathbf{k}_1\rangle, \dots, |\mathbf{k}_N\rangle\}$ with

$$\hat{H}_0 |\mathbf{k}_j\rangle = E_j |\mathbf{k}_j\rangle, \quad E_j = \frac{\hbar^2 k_j^2}{2m}.$$

Assume the following potential

$$\hat{V} = \lambda |\chi\rangle \langle\chi|, \quad |\chi\rangle = \sum_{j=1}^N c_j |\mathbf{k}_j\rangle.$$

- (i) Write the LS equation

$$|\psi^{(+)}\rangle = |\mathbf{k}_i\rangle + \hat{G}(E_i) \hat{V} |\psi^{(+)}\rangle, \quad \hat{G}(E_i) = \frac{1}{(E_i + i\epsilon)\mathbb{1} - \hat{H}_0},$$

in the momentum basis by defining components $\psi_j^{(+)} = \langle \mathbf{k}_j | \psi^{(+)} \rangle$. Show that

$$\psi_j^{(+)} = \delta_{j,i} + u_j \lambda \sum_{\ell=1}^N c_j c_{\ell}^* \psi_{\ell}^{(+)}, \quad u_j = \frac{1}{E_i - E_j + i\epsilon}.$$

- (ii) Introduce the collective amplitude

$$S \equiv \sum_{\ell=1}^N c_{\ell}^* \psi_{\ell}^{(+)}$$

and use the equation from (i) to show that S obeys

$$S = c_i^* + \lambda S \sum_{\ell=1}^N |c_{\ell}|^2 u_{\ell}.$$

(iii) Solve the equation from (ii) for S and insert the result back into $\psi_j^{(+)}$ to obtain an explicit expression

$$\psi_j^{(+)} = \delta_{j,i} + u_j \lambda c_j \frac{c_i^*}{1 - \lambda \sum_{\ell=1}^N |c_\ell|^2 u_\ell}.$$

(iv) (Connection to the δ -potential.) Specialize to a discretized δ -potential by choosing all coefficients equal, $c_j = 1/\sqrt{L^3}$, and show that the structure of $\psi_j^{(+)}$ reduces to the pattern discussed in the lecture for the 3D δ -potential in a finite momentum basis.

3. First Born approximation and Fourier transform of the potential

We consider the first Born approximation, where the exact T -matrix is replaced by the potential operator $\hat{V}$ such that

$$f^{(1)}(\theta, \phi) = -\frac{2m}{\hbar^2} \frac{L^{3/2}}{4\pi} \langle \mathbf{k}_f | \hat{V} | \mathbf{k}_i \rangle,$$

together with the box-normalized plane waves $\langle \mathbf{r} | \mathbf{k} \rangle = \frac{1}{L^{3/2}} e^{i\mathbf{k} \cdot \mathbf{r}}$.

(i) Insert position completeness relations to prove

$$\langle \mathbf{k}_f | \hat{V} | \mathbf{k}_i \rangle = \frac{1}{L^3} \int d^3r e^{-i(\mathbf{k}_f - \mathbf{k}_i) \cdot \mathbf{r}} V(\mathbf{r}).$$

(ii) Define the momentum transfer $\mathbf{q} = \mathbf{k}_f - \mathbf{k}_i$ and the Fourier transform of the potential

$$\tilde{V}(\mathbf{q}) = \int d^3r e^{-i\mathbf{q} \cdot \mathbf{r}} V(\mathbf{r}).$$

Rewrite $f^{(1)}(\theta, \phi)$ purely in terms of $\tilde{V}(\mathbf{q})$ and constants (up to overall box-normalization factors).

(iii) Specialize to an isotropic Gaussian potential

$$V(\mathbf{r}) = V_0 e^{-r^2/a^2}.$$

Compute $\tilde{V}(\mathbf{q})$ analytically by evaluating the 3D Gaussian integral, and obtain a closed expression for $f^{(1)}(\theta)$ for elastic scattering with $|\mathbf{k}_f| = |\mathbf{k}_i| = k$ so that

$$q = |\mathbf{q}| = 2k \sin \frac{\theta}{2}.$$

(iv) Discuss qualitatively how the angular distribution

$$\frac{d\sigma}{d\Omega} = |f^{(1)}(\theta)|^2$$

changes when:

- (a) you fix k and increase the range a ,
- (b) you fix a and increase the incoming momentum k .

Connect your discussion to the idea of diffraction from an extended object (wide vs. narrow angular patterns).